Why Ethics?

Integrated Ethics Team

2021-07-21

Student Materials

- web
- PDF

Overview

Ethics background required: None. This lab motivates the need for ethics education in computer science and/or data science.

Subject matter referred to in this lab: What do computer scientists/data scientists do?

Placement in overall ethics curriculum:

- Ideally, this is the first semester of the freshman year. We recommend that the students are 3-4 weeks into the first semester when this lab is administered.
- Recommended previous labs: None
- Recommended follow-up labs: Virtue Ethics, but any lab can follow

Time required:

- Out of class: none
- In class: 25 minutes

Learning objectives:

- Students gain a better understanding of how computer scientists and/or data scientists contribute to the world.
- Students have the opportunity to think about the responsibility that rests on their shoulders should they pursue a career in computer and/or data science.
- Students see the need for ethics education in computer and/or data science.
- Students understand what the study of ethics is

Ethical issues to be considered: This lab briefly considers many ethical issues in computing and data science. The actual issues covered will depend on the professor and class contributions.

Flow

- Students are introduced to the variety of contributions that Computer and Data Scientists make to the world.
- Student groups consider what could go wrong in the areas of Data and Computer Science that they just learned about.
- Student groups consider the good that could be done with products and analysis done in the best way.
- Ethics as a field of study is defined.
- Explanation that future labs will help provide tools for and give students practice in recognizing and making decisions about potential ethical issues that they could encounter in their future work.

Preparation: Read through the entire lab. Make copies of the job descriptions for computer and data scientists. Alternatively, make the descriptions available online.

Guide for Instructors

Lesson plan

Introduction (10 min)

Hand out (or point students to) the list below of jobs that computer and data scientists might be hired to do. Give them a moment to look it over, and ask if they would like to add anything.

Computer Science

- Software developer
  - Games
  - Social media
  - Airplane/Car navigation systems
  - Predictive algorithms (who is a good candidate for a loan or job, predicting future weather or erosion patterns, predicting if a person who is arrested is a flight risk)
  - Science applications (finding genetically-related diseases, analyzing tumor progression/reduction, drug discovery)
  - Compilers/operating systems

- Database Administrator
  - Design systems to store and retrieve information
- Computer hardware engineer
  - Design the next phone, computer or tablet
  - Create/test circuit boards, routers, memory devices
  - Create ways for disabled to move and/or communicate
- Information security analyst
  - Detect cyber attacks
  - Prevent cyber attacks
- Web Developer
  - Create the technical structure for websites
  - Make sure that web pages are accessible and easily downloadable
  - Structure sites to maximize the number of page views and visitors through search engine optimization.

Data Science

- Data Analytics
  - Help develop strategies to make better business decisions
  - Create/use programs to help individuals invest money
  - Track and analyze website usage patterns
- Data architect
  - Find, clean and organize data
  - Look for patterns in data collected
- Machine learning
  - Develop predictive algorithms based on what has happened in the past (medicine, business, job/student success, economy, climate change, disease spread etc.)
- Statistician
  - Interpret, analyze, and report statistical information (could be related to research in any field: business, psychology, sociology, medicine..)

Activity (15 min)

Read or summarize to students: It should be apparent from the list that computer and data scientists can hold jobs that are very influential in society. With such power, people in these positions can bring great good, or, if they are not careful, great harm. In the next 10 minutes do the following in your group:

- Choose a note-taker and spokesperson so that your ideas can be shared with the larger group.
- Brainstorm with your group things that could go wrong in any of these areas if they are not handled carefully.
- Brainstorm with your group the things that could be beneficial to society if they were handled in the best possible way.

Reflection as a class (10 min)

After 10 minutes, return the students to the larger group to share their thoughts. See if an individual can summarize the need to consider carefully how products are designed and data analyzed based on the class discussion.

Here are some suggestions if the students are unable to come up with ideas:

Things that can go wrong:

- Games can be violent and/or addictive
- Social media can also be addictive and also divisive. It can propagate wrong ideas and bullying, can enable predators, and may trivialize or commoditize relationships.
- Security specialists could use their training for wrong purposes.
- Hardware engineers could produce too much e-waste and/or use too much energy.
- Statisticians could skew results to force desired outcomes.
- Data architects could clean data in a way that eliminates diversity from the data.
- Web designers can embed software to collect unauthorized data.

Things that can make a positive difference:

- Games can build friendships and provide opportunities to relax and de-stress
- Social media can keep people in touch with each other
- Security specialists can protect privacy
- Hardware engineers can provide devices that help the physically challenged and elderly to be more secure, productive, and connected.

- Statisticians could interpret data is a way that helps make informed decisions on healthcare or other policy
- Web designers could create portals for information with accessibility features that improve the lives of many.

Wrap up (Read or give to students)

According to author Sara Baase, in her book A Gift of Fire [1], "Ethics is the study of what it means to do the right thing." While this is a simple definition, it turns out that the study of ethics can be a complex topic. However, it is certainly one that computer and data scientists, because of the nature of their work, need to carefully consider as they are trained and then move forward into their careers.

If your university has a plan to use future labs, mention that students will have opportunities in other classes to learn tools to help them as computer/data scientists make ethical decisions on implementations and to anticipate detrimental outcomes.

Assessment

1. Name 3 jobs that you as a computer or data scientist could hold in the future
2. What is ethics?
3. Give an example of a product design or a data analysis task that could be have positive effects or negative effects on its users depending on how the data scientist, computer scientist, or engineer approached the implementation.

Online teaching

This lesson is easily adapted to an asynchronous online environment where a student would complete the assignment individually and then post some of their ideas while also responding to the ideas of others.

References

Bibliography

[1] S. Baase, A gift of fire. Pearson Education Limited, 2012.